Electron Charge-to-Mass Ratio: Thomson's Experiment

Electron Charge to Mass Ratio Lab · oPhysics

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics

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Simulation: https://ophysics.com/em2a.html
Curriculum links: quantum physics · atomic structure

Learning intentions

- I can describe how electric and magnetic fields were used to determine properties of the electron.
- I can apply circular motion in a magnetic field to calculate a charge-to-mass ratio.

Getting started

Open the simulation and work through each step, recording the measurements it asks you to take.

Tasks

1. Follow the simulation's steps and record the measurements it provides (accelerating voltage, magnetic field strength, and radius of curvature).

2. Using your recorded measurements, calculate the electron's charge-to-mass ratio, e/m.

Working:

3. Compare your calculated e/m to the accepted value of 1.76×10^{11} C/kg. Calculate your percentage error.

4. Identify two possible sources of error in this simplified simulation that would not occur in the same way in a real vacuum-tube experiment.

Extension (optional)

Explain why measuring e/m (rather than e or m individually) was the key first step toward discovering the electron, and why Millikan's later oil-drop experiment (finding e alone) was needed to complete the picture.